INCH-POUND

MIL-L-40085F
27 March 1989
SUPERSEDING
MIL-L-40085E
31 December 1985

MILITARY SPECIFICATION

* LINK ASSEMBLY, TYPE IV, AIR DELIVERY

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document covers one type of link assembly for single suspension or extraction airdrop rigging.

2. APPLICABLE DOCUMENTS

* 2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

PPP-B-566 - Boxes, Folding, Paperboard
PPP-B-601 - Boxes, Wood, Cleated-Plywood

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Natick Research, Development, and Engineering Center, Natick, MA 01760-5014 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 1670

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PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner
PPP-B-676 - Boxes, Setup
PPP-T-60 - Tape, Packaging, Waterproof
PPP-T-76 - Tape, Packaging, Paper (For Carton Sealing)

MILITARY

MIL-H-6875 - Heat Treatment of Steel, Process For

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-147 - Palletized Unit Loads

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drqwings, and publications form a part of this document to the extent specified herein. Unless otherwixe specified, the issues are those cited in the solicitation.

DRAWINGS

U.S. ARMY NATICK RESEARCH, DEVELOPMENT, AND ENGINEERING CENTER

11-1-3359 - Link, Single Suspension or Extraction, Air Delivery Assembly
11-1-3360 - Body, Assembly
11-1-3361 - Body
11-1-3362 - Pin
11-1-3363 - Roller
11-1-3364 - Spring
11-1-3365 - Button
11-1-3366 - Plate
11-1-3367 - Lock

(Copies of drawings required by contractors are available from the U.S. Army Natick Research, Development, and Engineering Center, Natick, MA 01760-5014.)

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2.2 Non-Government publications. The following document forms a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 3951 - Standard Practice for Commercial Packaging

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1187.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes preedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. RETIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.3.

3.2 Materials and components. Materials and components shall be as specified herein and on drawings 11-1-3359 through 11-1-3367 (see 2.1.2), as applicable. Materials not definitely specified shall be of the quality normally used by the manufacturer, provided that the -completed item complies with all provisions of this document. It is encouraed that recycled material be used when practical as long as it meets the requirements of this document.

3.3 Performance. When tested as specified in 4.4.5, the link assembly shall show no evidence of cracks, loose pins, or pins pulled away from the body so that the end of the pins are not flush with the outside surface of the plate, or permanent deformation (lock and plate cannot be removed or reassembled on pins without use of tools).

3.4 Function. The link assembly shall operate freely. With the lock plate in the locked positon, the button shall be easily depressed by hand to clear the lock plated, and the lock plate shall freely return to lock position when the button is released. When the button is depressed, the lock plate shall be movable by hand to the unlocked positon. With the lock plate in the unlocked position, the removable plate shall be easily removable and replaceable by hand. The spools shall be easily removable and replaceable by hand and shall not bind on the pins.

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3.5 Identificaiton marking. Marking for identification shall be as specified on the applicable drawings.

3.6 Workmanship. The end item shall conform to the quality of product established by this document and the occurrence of defects shall not exceed the applicable acceptable quality levels.

* 4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 Responsibility for dimensional requirements. Unless otherwise specified in the contract or purchase order, the contractor is responsible for ensuring that all specified dimensions have been met. When dimensions cannot be examined on the end item, inspection shall be made at any point, or at all. points in the manufacturing process necessary to assure compliance with all dimensional requirements.

4.1.3 Certificates of compliance. When certificates of compliance are submitted, the Government reserves the right to inspect such items to determine the validity of the certification.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).

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4.3 First article inspection. When a first article is required (see 3.1 and 6.2) , it shall be examined for the defects specified in 4.4.3, 4.4.4, 4.4.5, and 4.4.6.

4.4 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.4.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase" document.

4.4.2 In-process inspection. Inspection of subassemblies shall be made to ascertain that construction on details which cannot be examined in the finished product are in accordance with specified requirements. The Government reserves the right to exclude from consideration for acceptance, any material or service for which in-process inspection has indicated nonconformance.

4.4.2.1 Heat treatment examination. Heat treated components shall be inspected to determine compliance with requirements specified on the applicble drawings. The inspection shall be in accordance with MIL-H-6875 to include hardness test, except that the inspection level shall be S-3 and the acceptable quality leveel (AQL), expressed in terms of defects per hundred Units, shall be 1.0.

4.4.2.2 Inspection of force fit dimension. Prior to assembly, components shown on Drawing 11-1-3360 shall be inspected to determine compliance with the specified force fit dimensions and assembly requirements. The lot size shall be expressed in individual components. The sample unit shall be one pin and one body, as applicable. The inspection level shall be III and the AQL, expressed in terms of defects per hundred units, shall be 0.10.

4.4.3 End item visual examination. The end items shall be examined for the defects listed in table I. The lot size shall be expressed in units of link assemblies. The sample unit shall be one link assembly. The inspection level shall be II and the AQL, expressed in terms of defects per hundred units, shall be 1.5 for major defects and 6.5 for total defects.

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TABLE I. End item visual defects

Examine	Defect	Classification Major	Minor
Finish, protective	Any breaks through coating to base metal		201
	Any component with primary finish not as specified	101	
	Any area of rust or corrosion	102	
	Dirt, oil, grease, or other foreign matter		202
	Supplementary finish omitted or not applied as specified		203
Finish, surface	Any surface roughness exceeding specified limit	103	
Construction and workmanship	Any component missing	104	
	Any component cracked or fractured	105	
	Any sharp edges or burrs	106	
	Any operation omitted or not performed as specified	107	
	Any component not assembled or joined as specified	108	
Identification marking	Missing, illegible, incomplete, incorrect, or not permanent		204

4.4.4 End item dimensional examination. The end items shall be examined to determine compliance with the dimensions specified on the applicable drawings (excluding the separate force fit dimensions and assembly in 4.4.2.2). The lot size shall be expressed in units of link assemblies. The sample unit shall be one link assembly. Any deviations therefrom shall constitute a defect. The inspection level shall be S-3 and the AQL, expressed in term of defects per hundred units, shall be 6.5.

4.4.5 End item proofload and side load examination. The link assemblies shall be subjected to a separate static proofload test of 40,000 pounds and side load test of 2,000 pounds as shown on Drawing 11-1-3359 to determine compliance wiht 3.3. The load shall be distributed across the full width of the rollers or plates, as applicable. The sample unit shall be one fully assembled link. Any nonconformance shall constitute a defect. The inspection level shall be S-3 and the AQL, expressed in terms of defects per hundred units, shall be 1.0.

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4.4.6 End item function examination. Each link assembly shall be examined to determine compliance with 3.4. Any of the following link defects shall be cause for rejection of the link assembly:

a. With lock plate in locked position, button cannot be depressed by hand to clear the lock plate, or does not freely return to lock position when released.

b. With button in depressed position, lock plate cannot be moved by hand to unlocked position.

c. With lock plate in unlocked position, removable plate cannot be removed and replaced by hand.

d. Spools bind on pins or cannot be removed and replaced by hand.

4.7 Packaging examination. The fully packaged end items shall be examined for the defects listed below. The lot size shall be expressed in units of shipping containers. The sample unit shall be one shipping contain fully packaged. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 2.5.

Examine	Defect
Marking (exterior and interior)	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application
Materials	Any component missing, damaged, or not as specified
Workmanship	Inadequate application of components, such as incomplete sealing or closure of flap, improper taping, loose strapping, or inadequate stapling Bulged or distorted container
Content	Number per container is more or less than required

4.4.8 Palletization examination. The fully packaged and palletized end items shall be examined for the defects listed below. The lot size shall be expressed in units of palletized unit loads. The sample unit shall be one palletized unit load, fully packaged. The inspection level shall be S-1 and the AQL, expressed in terms of defects per hundred units, shall be 6.5.

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Examine	Defect
Finished dimensions	Length, width, or height exceeds specified maximum requirement
Palletization	Pallet pattern not as specified Interlocking of loads not as specified Load not bonded as specified
Weight	Exceeds maximum load limits
Marking	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application

5. 5. PACKAGING

* 5.1 Preservaiton. Preservation shall be level A or Commercial., as specified (see 6.2).

5.1.1 Level A. Each link assembly shall be unit packed in a paperboard box conforming to variety 2, style III, type G, class i of PPP-B-566, or type I, variety 2, class A, style 4 of PPP-B-676. Inside dimensions of each box shall be 5 inches in length, 3 inches in width, and 2 inches in depth. Each box shall be securely closed with 1-inch minimum width pressure sensitive tape conforming to PPP-T-60 or PPP-T-76, applied at the center of the length opening and along the bottom and up each side at least 1 inch.

5.1.2 Commercial. Each link assembly shall be preserved in accordance with ASTM D 3951.

5.2 Packing. Packing shall be level A, B, or Commercial as specified (see 6.2).

5.2.1 Level A packing. Sixty link assemblies, preserved as specified in 5.1, shall be packed in a shipping container conforming to overseas type, style A or I, grade A, type 2 load of PPP-B-601, or class 2, style 2 or 4, grade A or B, type 2 load of PPP-B-621, as specified. Level A unit packs shall be packed on their sides, four in length, five in width, and three in depth within a shipping container. Inside dimensions of each shipping container shall be approximately 21-1/2 inches in length, 11-1/4 inches in width, and 9-3/4 inches in depth. Approximate dimensions are furnished as guide only. Each shipping container shall be closed and reinforced in accordance with the appendix of the applicable container document.

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5.2.2 Level B packing. Sixty link assemblies, preserved as specified in 5.1, shall be packed in a shipping container conforming to domestic type, style A or I, grade B, type 2 load of PPP-B-601, or class 1, style 2 or 4, grade B, type 2 load of PPP-B-621, as specified. Level A unit packs shall be packed on their sides, four in length, five in width, and three in depth within a shipping container. Inside dimensions of each shipping container shall be approximately 21-1/2 inches in length, 11-1/4 inches in width, and 9-3/4 inches in depth. Approximate dimensions are furnished as a guide only. Each shipping container shall be closed in accordance with the appendix of the applicable container document.

5.2.3 Commercial packing. Link assemblies, preserved as specified in 5.1, shall be packed in accordance with ASTM D 3951.

5.3 Palletization. When specified (see 6.2), link assemblies, packed as specified in 5.2.2 or 5.2.3, shall be palletized on a 4-way entry pallet in accordance with load type Ia of MIL-STD-147. Pallet type shall be type I (4-way entry), type IV or type V in accordance with MIL-STD-147. Each prepared load shall be bonded with primary and secondary straps in accordance with bonding means K and L or film bonding means 0 or P. Pallet pattern shall be number 35 in accordance with the appendix of MIL-STD-147. Interlocking of loads shall be effected by reversing the pattern of each course.

5.4 Marking. In addition to any special marking required by the contract or pure-der, shipping containers shall be marked in accordance with MIL-STD-129 or ASTM D 3951, as applicable.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The link assembly is used in attaching slings and risers connecting the suspension and extraction group assemblies of heavy-drop parachute kits.

6.2 Axquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this document.
b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2-2).
c. When a first article is required (see 3.1, 4.3, and 6.3).
d. Levels of preservation and packing (see 5.1 and 5.2).
e. When palletization is required (see 5.3).

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6.3 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of FAR 52.209. The first article should be a preproduction sample. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for selection, inspection, and approval of the first article.

6.4 Subject term (key word) listing.

Assembly, suspension
Parachutes
Rigging, air drop

6.5 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any ibaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:

Army - GL
Navy - AS
Air Force - 99

Review activities:

Army - AV
Navy - MC
Air Force - 11

Preparing activity:

Army - GL

(Project 1670-0766)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-L-40085F	Link Assembly, Type IV, Air Delivery

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

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